Editorial

ANNALS OF MEDICINE & SURGERY

OPEN

# FDA approval of Casgevy and Lyfgenia: a dual breakthrough in gene therapies for sickle cell disease

Zainab Syyeda Rahmat, MBBS[a], Mohammad Haris Ali, MBBS[b], Muhammad Talha, MBBS[b], Md. Al Hasibuzzaman, MBBS[c,*]

## Introduction

On 8 December 2023, the US Food and Drug Administration (FDA) granted approval to two cell-based gene therapies, Casgevy (Exagamglogene autotemcel) and Lyfgenia (Lovotibeglogene autotemcel), for the treatment of sickle cell disease (SCD) in patients aged 12 years and older. This marks a significant advancement in the treatment of this condition[1]. Notably, Casgevy is the first gene therapy employing CRISPR/Cas9 gene editing technology to receive FDA approval, not only for SCD but also for beta-thalassemia, another inherited blood disorder. The integration of Casgevy's therapeutic framework represents a revolutionary advancement in the precision and efficacy of gene editing techniques applied clinically. Current treatments for SCD include hydroxycarbamide, L-glutamine, and blood transfusions, each with limitations and varying effectiveness. Bone marrow transplantation (BMT) is the sole curative option but is reserved for severe cases due to its complexity and risks[2].

These approvals signal a new era in biomedical innovation, where gene therapies are envisioned not only as potential cures but also as transformative therapies capable of addressing the underlying genetic causes of diseases previously considered untreatable by medication alone. The regulatory endorsement of Casgevy, in particular, highlights the FDA's recognition of CRISPR/Cas9 technology's safety and efficacy in clinical applications, setting a precedent for future developments in gene editing therapies across a spectrum of genetic disorders[1].

[a]Faculty of Medicine, Dow Medical College, Dow University of Health Sciences, Karachi, [b]Shaikh Khalifa Bin Zayed Al-Nahyan Medical and Dental College, Lahore, Pakistan and [c]Institute of Nutrition and Food Science, University of Dhaka, Dhaka, Bangladesh

Sponsorships or competing interests that may be relevant to content are disclosed at the end of this article.

*Corresponding author. Address: University of Dhaka, Dhaka, Bangladesh, Dhaka 1000. Tel.: +880 172 320 2217; fax: +880 2966 7222.
E-mail: al.hasibuzzaman.hasib@gmail.com (M. A. Hasibuzzaman).

Copyright © 2024 The Author(s). Published by Wolters Kluwer Health, Inc. This is an open access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

Annals of Medicine & Surgery (2024) 86:4966–4968

Received 1 May 2024; Accepted 16 July 2024

Published online 25 July 2024

http://dx.doi.org/10.1097/MS9.0000000000002409

## Methods

Articles on PubMed/Medline were searched and reviewed thoroughly for trials on the respective cell therapies and SCD until April 2024. Relevant and important information was screened and included in this study.

## Discussion

SCD is a genetic disorder caused by a mutation in the beta-globin gene on chromosome 11, where the gene sequence GAG is changed to GTG. This mutation leads to the substitution of the amino acid valine for glutamic acid in the β-globin chain, resulting in the formation of hemoglobin S (HbS) instead of hemoglobin A (HbA). Approximately 300 000 infants are born with this condition every year globally, with the number projected to increase to 400 000 by the year 2050[3]. It is estimated that ~100 000 Americans suffer from this condition, emphasizing the need for a novel approach[4].

The pathophysiology of SCD involves the abnormal behavior of hemoglobin in a deoxygenated environment, where polymerization of HbS causes red blood cells (RBCs) to assume a sickle shape and become rigid, making them susceptible to excessive breakdown and destruction (hemolysis). The inflexible RBCs disrupt blood flow and compromise endothelial wall integrity, leading to recurrent vaso-occlusive episodes (VOEs) characterized by severe pain, edema, necrosis, and organ damage. Additionally, due to excessive hemolysis, the lifespan of RBCs is shortened to 10–20 days compared to the usual 120 days, contributing to significant anemia[5]. Other complications include acute aplastic crisis, infection, splenic sequestration crisis, and psychosocial impact[6].

The current treatment strategy for SCD includes the use of the medication hydroxycarbamide (hydroxyurea), which increases the production of fetal hemoglobin (HbF) as elevated levels of HbF are associated with reduced mortality and morbidity in these patients. However, there is uncertainty about its long-term use, with indications varying based on phenotype, age, and individual practices[7]. Other treatment options, such as L-Glutamine and blood transfusion, exist, each with its own set of drawbacks. Bone marrow transplantation (BMT) stands as the sole cure for SCD, offering a high survival rate, but it is typically reserved for severe cases. Thus, these limited therapies indicate the need for a more effective and reliable cure.

Casgevy, the first CRISPR/Cas9-approved gene therapy, treats SCD in patients aged 12 and above with recurrent vaso-occlusive crises (VOCs). The therapy works by ex-vivo gene editing at the

erythroid enhancer region of the BCL11A gene in the patient's own CD34 hematopoietic stem and progenitor cells (HSPCs). BCL11A is a transcription factor that contributes to the suppression of HbF production after birth[8]. Cas9, a programmable nuclease, induces double strand breaks at the said enhancer region, which activates the DNA repair mechanism. The process leads to the generation of insertions or deletions, resulting in the inactivation of the BCL11A gene[9]. Hence, these edited cells increase HbF production, preventing the sickling of RBCs and improving oxygen delivery, reducing the mortality and morbidity associated with SCD[10].

On the other hand, Lyfgenia utilizes a lentiviral vector as its gene delivery mechanism, genetically altering the patient's blood stem cells via lentiviral addition of βT87Q-globin, a modified beta-globin, in HSPCs to generate HbAT87Q. HbAT87Q has functional similarities to hemoglobin A, has an anti-sickling effect, and allows for precise quantification of vector-derived therapeutic globin expression in vivo[11].

The FDA approval comes after positive results were seen in the clinical trials of these therapies in patients with SCD. The safety and efficacy of Casgevy are being assessed in an ongoing (single-arm, open-label, multi-center, single-dose) Phase 1/2 trial involving adult and adolescent patients with a documented SCD genotype. The enrolled patients had a history of at least two severe VOCs within the two years before screening. Out of 44 patients treated, 29 (93.5%) were free from severe episodes of VOC for a minimum of 12 consecutive months within the 24-month follow-up period. Means of HbF and Total Hb were recorded and analyzed, showing that every patient achieved successful engraftment, and there were no occurrences of graft rejections during follow-ups ranging from 4.9 to 22.4 months[12].

In the case of Lyfgenia, its safety and effectiveness were established through a 24-month (non-randomized, open-label, multi-site, single-dose) Phase 1/2 multi-center study involving diagnosed SCD patients with severe conditions. The patients were aged 12–50 with a history of recurrent VOCs. The evaluation focused on achieving complete resolution of VOEs (VOE-CR) between 6 and 18 months post-Lyfgenia infusion. The analysis indicated that out of 32 patients, 28 (88%) attained VOE-CR during this period, with follow-ups ranging from 8.5 to 28.5 months[13].

Some of the common side effects in both trials include mouth sores, low levels of platelets and white blood cells, and febrile neutropenia. Nausea, musculoskeletal pain, abdominal pain, vomiting, headache, and itching were additional side effects seen in patients given Casgevy. Even though no hematologic malignancy was observed in the trials, such malignancies have been recorded in patients treated with Lyfgenia, and hence, patients receiving this therapy must be closely monitored[14].

These therapies offer significant potential to transform medicine but introduce substantial ethical issues, particularly around the editing of the human germline, which would pass changes to future generations. Researchers and bioethicists agree that germline editing for reproductive purposes should be postponed until further research confirms its safety and effectiveness. The high costs of treatments such as Casgevy and Lyfgenia raise concerns about societal equity and justice, necessitating robust ethical and legal regulations. Key safety issues, including the risks of mosaicism and off-target effects, require detailed examination, with an emphasis on long-term safety and efficacy studies and the evaluation of combination therapies for diseases like SCD. Addressing these ethical challenges demands continuous interdisciplinary collaboration, ethical oversight, and the creation of policies to ensure responsible research and application[15].

Casgevy and Lyfgenia have been hailed as breakthroughs in the field of medicine, providing a potential cure for hundreds of thousands of patients affected by SCD. Both therapies have been approved by the FDA after showing efficacy in separate clinical trials. Until now, no effective treatment has been discovered for SCD except bone marrow transplantation, which contains its own set of risks. Both Casgevy and Lyfgenia hold promising futures for curing SCD in the United States and potentially worldwide. These advancements are poised to accelerate research and development in gene therapy, fostering a burgeoning field dedicated to enhancing treatment outcomes and quality of life for patients with genetic disorders worldwide. This noteworthy development underscores the promise of cell-based therapies as a viable alternative to currently prescribed standard drugs.

## Ethical approval

Institutional IRB approval was not needed for this article.

## Consent

All authors consented for approval.

## Source of funding

The authors report no external funding.

## Author contribution

Z.S.R.: conceptualization, methodology, validation, writing—original draft preparation, writing—reviewing and editing. M.H.A.: methodology, validation, writing—original draft preparation, writing—reviewing and editing. M.T.: writing—reviewing and editing, investigation. M.A.H.: writing—reviewing and editing, submission.

## Conflicts of interest disclosure

The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest. The authors declare that they have no conflicts of interest.

## Research registration unique identifying number (UIN)

Not applicable.

## Guarantor

Mohammad Haris Ali.

## Data availability statement

None.

## Provenance and peer review

Not applicable.

## References


[1] FDA News Release. 8 Dec 2023. FDA Approves First Gene Therapies to Treat Patients with Sickle Cell Disease. U.S. Food and Drug Administration. FDA. https://www.fda.gov/news-events/press-announcements/fda-approves-first-gene-therapies-treat-patients-sickle-cell-disease

[2] Leibovitch JN, Tambe AV, Cimpeanu E, *et al*. l-glutamine, crizanlizumab, voxelotor, and cell-based therapy for adult sickle cell disease: hype or hope? Blood Rev 2022;53:100925.

[3] Piel FB, Hay SI, Gupta S, *et al*. Global burden of sickle cell anaemia in children under five, 2010–2050: modelling based on demographics, excess mortality, and interventions. PLoS Med 2013;10:e1001484.

[4] Sickle Cell Disease. 15 May 2024. Data and Statistics on Sickle Cell Disease. U.S. Centers for Disease Control and Prevention, CDC. https://www.cdc.gov/sickle-cell/data/index.html

[5] Sebastiani P, Nolan VG, Baldwin CT, *et al*. A network model to predict the risk of death in sickle cell disease. Blood 2007;110:2727–35.

[6] Tezol O, Karahan F, Unal S. Sickle cell disease and psychosocial well-being: comparison of patients with preclinical and clinical avascular necrosis of the femoral head. Turk Arch Pediatr 2021;56: 308–15; Published 2021 Jul 1.

[7] Ware RE. Optimizing hydroxyurea therapy for sickle cell anemia. Hematol Am Soc Hematol Educ Program 2015;2015:436–43.

[8] Bauer DE, Orkin SH. Hemoglobin switching's surprise: the versatile transcription factor BCL11A is a master repressor of fetal hemoglobin. Curr Opin Genet Dev 2015;33:62–70.

[9] Onimoe G, Rotz S. Sickle cell disease: a primary care update. Cleve Clin J Med 2020;87:19–27.

[10] Demirci S, Leonard A, Haro-Mora JJ, *et al*. CRISPR/Cas9 for Sickle Cell Disease: Applications, Future Possibilities, and Challenges. In: Turksen K, ed. Cell Biology and Translational Medicine, Volume 5: Stem Cells: Translational Science to Therapy Advances in Experimental Medicine and Biology. Springer International Publishing; 2019:pp. 37–52.

[11] Demirci S, Gudmundsdottir B, Li Q, *et al*. βT87Q-globin gene therapy reduces sickle hemoglobin production, allowing for ex vivo anti-sickling activity in human erythroid cells. Mol Ther Methods Clin Dev 2020;17: 912–21.

[12] A safety and efficacy study evaluating CTX001 in subjects with severe sickle cell disease. (n.d.). Vertex Clinical Trials. https://clinicaltrials.vrtx.com/safety-and-efficacy-study-evaluating-ctx001-subjects-severe-sickle-cell-disease

[13] A Study Evaluating Gene Therapy With BB305 Lentiviral Vector in Sickle Cell Disease. (n.d.) bluebird bio. https://clinicaltrials.gov/study/NCT04293185

[14] FDA News Release. 19 December, 2023. FDA Roundup: December 19, 2023. U.S. Food and Drug Administration. FDA. https://www.fda.gov/news-events/press-announcements/fda-roundup-december-19-2023

[15] Orkin SH. Recent advances in globin research using genome-wide association studies and gene editing. Ann N Y Acad Sci 2016;1368:5–10.